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**Asset Bubble Detection: An Econometric
Approach**

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*To all the Iranian heroes who lost their lives
in pursuit of freedom.*

Abstract

In the past decade, we have witnessed a remarkable capital concentration in the technology sector which was heavily influenced by the rapid commercialization of generative Artificial Intelligence. To determine whether the price movements of mega-cap equities known as the "Magnificent Seven" represent justified fundamental growth or idiosyncratic speculative bubbles remains as an important challenge for quantitative risk management. Traditional unit root tests fail to detect periodically collapsing exuberance, and also applying explosive tests to nominal prices causes false positives.

This thesis converts nominal prices to relative Price-to-Index ratios to neutralize aggregate market beta. The Generalized Supremum Augmented Dickey-Fuller (GSADF) methodology proposed by P. C. Phillips, Shi, and Yu (2015) is then applied to these metrics to date-stamp explosive market regimes. Moreover, this research uses near-unit-root limit theory introduced by P. C. Phillips (2023). By using Cauchy asymptotics to estimate the localizing rate (γ), the methodology measures the magnitude of the exuberance which then would categorize the bubbles as Mildly Integrated, Local-to-Unity, or Mildly Explosive.

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Chapter 1

Introduction

1.1 Background and Motivation

The fundamental challenge in quantitative asset pricing is to understand whether an expansion of value is driven by economic growth or a speculative bubble. Due to different innovations - such as AI - over the past decade, equity markets specifically the technology sector have experienced remarkable capital accumulation. The valuation of technology equities has tested the boundaries of traditional financial models ever since.

At the center of these changes are the "Magnificent Seven" and the Nasdaq-100 index. Asset prices have increased aggressively because of these capital flows into firms building advanced semiconductor and large language models. However, the main challenge is identifying whether these price movements shows a fundamental repricing of technological dominance (Pástor and Veronesi, 2009) or a speculative frenzy. To mathematically be able to detect and measure the severity of these explosive market regimes, especially for risk management, is of importance.

1.2 Problem Statement

Identifying a speculative bubble empirically, presents a big econometric challenge. Traditional unit root tests, like the standard Augmented Dickey-Fuller (ADF) test, fail to detect periodically collapsing bubbles because of market crashes that happen in the post-bubble period, commonly referred to as the Evans Critique (Evans, 1991). In addition, applying autoregressive tests to the nominal prices of stocks causes a high risk of false positives. If the underlying cause of a surge in the asset price are the market factors or the fundamentals of a company, the stock is appreciating without exhibiting an isolated, idiosyncratic bubble.

When the explosive regime is isolated, the second challenge of standard time-series econometrics would be presented. When an asset is in a speculative phase, its autoregressive coefficient exceeds unity ($\theta > 1$). Gaussian limit theory is invalid in this non-stationary state, therefore the confidence intervals and standard errors would be mathematically invalid which would prevent accurately measuring the severity of the exuberance.

1.3 Research Objective

The aim of this thesis is to resolve these challenges by using a robust econometric framework to detect, isolate and measure speculative bubbles in the technology sector. The research is guided by three primary objectives:

1. **Isolation of Idiosyncratic Exuberance:** To neutralize any broader impact from sector-wide beta by transforming prices into relative Price-to-Index ratios (e.g., individual equity prices scaled by the S&P500 or Nasdaq-100).
2. **Bubble Detection:** Applying the methodology of P. C. Phillips, Shi, and Yu (2015) to these relative ratios and date-stamping the beginning and the end of a speculative regime.
3. **Measuring Bubble Severity:** Using near-unit-root limit theory introduced by P. C. Phillips (2023). This thesis will calculate the severity parameter (γ) to categorize the asset's regime as Mildly Integrated, Local-to-Unity, or Mildly Explosive.

1.4 Contribution to the Literature

This research contributes to the existing financial econometrics literature in two distinct ways. First, it refines the application of the PSY framework by strictly enforcing relative pricing metrics, thereby providing a cleaner, more rigorous isolation of idiosyncratic exuberance than models reliant on absolute price data. Second, it bridges the gap between theoretical econometrics and practical risk management by fully integrating the P. C. Phillips (2023) limit theory. Rather than settling for binary bubble detection, this thesis provides a quantifiable, asymmetric confidence interval for bubble severity, offering a highly advanced tool for assessing structural risk in the AI-driven equity market.

1.5 Structure of the Thesis

The structure of the thesis is organized as follows. Chapter 2 is a review of the relevant literature, including theoretical definitions of speculative bubbles, macroeconomic catalysts of the past decade, and the evolution of explosive limit theory. Chapter 3 contains the core methodology, defining the PSY framework, the construction of the Price-to-Index dataset, and the specific parameters required for Cauchy inference. Chapter 4 demonstrates the empirical results, applying the methodology to equities of the technology sector. At the end, Conclusion concludes the thesis summarizing the findings.

Chapter 2

Literature Review

2.1 Speculative Bubbles and Technological Revolutions

Speculative bubbles are deeply connected with periods of technological transformation. When disruptive innovations of massive scale occur, it creates an environment when capital accumulation can mean either a true shift in paradigm or an irrational exuberance. This section provides the foundation to understand these explosive market dynamics.

2.1.1 Asset Price Bubble and Fundamental Value

In financial economics, the market value of an asset is related to the present value of its future payoffs. This relationship is modeled via the present value identity, where the asset price (P_t) is calculated as the sum of its discounted expected payoffs—such as dividends or earnings—and an observable component (B_t) representing the bubble:

$$P_t = \sum_{i=0}^{\infty} \left(\frac{1}{1+r_f} \right)^i \mathbb{E}_t(D_{t+i} + U_{t+i}) + B_t \quad (2.1.1)$$

where r_f is the risk-free rate of return, D_{t+i} represents the observable payoff streams, and U_{t+i} is unobservable fundamental factors.

Under the null hypothesis of market efficiency and fundamental pricing, the initial condition of the component is zero ($B_0 = 0$), showing that asset prices are driven only by economic fundamentals, which generally exhibit at most integrated of order one ($I(1)$) non-stationary behavior.

However, when $B_0 > 0$, a speculative bubble emerges. To sustain its existence over time, this submartingale bubble component should satisfy the submartingale

property:

$$\mathbb{E}_t(B_{t+1}) = (1 + r_f)B_t \quad (2.1.2)$$

Since the discount factor $1/(1+r_f)$ is counterbalanced by the growth rate of the bubble component $(1 + r_f)$, the element B_t expands exponentially. As B_t grows, eventually it dominates the price dynamics of the asset, transmitting explosive or mildly explosive autoregressive behavior to the price series (P_t) and severing its connection to underlying fundamental cash flows.

2.1.2 Rational Expectations vs. Behavioral Finance

The academic literature is divided regarding the psychological and informational mechanisms that trigger this explosive divergence.

The Rational Expectations Framework: Initiated by theories such as Blanchard and Watson (1982), it argues that a bubble does not imply investor irrationality. Within a rational bubble framework, market participants are fully aware that an asset's price has shifted above its fundamental baseline. However, it is rational for an individual to purchase or hold the asset if they are compensated with a high probability of ongoing capital gains before the bubble bursts. This explosive trajectory is continued as a rational response to a self-fulfilling prophecy, where the expected appreciation of future price explains current overvaluation.

The Behavioral Finance Framework: In contrast, the behavioral perspective established by Shiller (2000) defines bubbles as manifestations of "irrational exuberance". Shiller describes speculative bubbles as a form of psychological contagion or social epidemic, where a positive feedback loop changes investor perceptions. Initial price gains encourage media attention and public enthusiasm, which increases investor greed and FOMO (Fear of Missing Out). This induces herd behavior, causing market participants to ignore private information and fundamental valuation metrics altogether.

2.1.3 Technological Paradigms and Valuation Uncertainty

While the debate between rational and behavioral drivers carries on, empirical consensus exhibits that speculative bubbles do not emerge by accident; they are paired with technological evolutions and structural economic paradigm shifts.

Technological innovations have characteristics that make them prime breeding grounds for financial exuberance. When a groundbreaking innovation—such as a new computing architecture, energy system or artificial intelligence—is introduced across the globe, it causes a period of continuous innovation and adoption. However, measuring the exact economic value of such transformations is very difficult.

Pástor and Veronesi (2009) formalize this relationship by showing that technological revolutions introduce fundamental uncertainty into financial markets. Since there is no data baseline to precisely forecast how a paradigm-shifting technology will translate into future cash flows, profit margins or macroeconomic productivity, traditional asset pricing models fail.

Historical experience provides evidence of this structural vulnerability:

The 1920s Radio and Automotive Boom: The introduction of radio broadcasting via Radio Corporation of America (RCA) transformed society. The huge enthusiasm about this technology caused RCA's stock price to increase from \$94.50 in early 1928 to \$505 in late 1929, causing its price-to-earnings (P/E) ratio to a peak of 60 right before the October 1929 market crash.

The 1990s Dotcom Bubble: The commercialization of internet-based technology led to optimism regarding a "New Economy". Infrastructure giants like Cisco experienced a massive overvaluation, with stock prices peaking in March 2000, before collapsing by almost 90% over the subsequent year.

The 2020s Artificial Intelligence Era: The massive increase in interest and implementation of large language models, generative AI applications, and specialized parallel-computation GPU chip technology represents the most recent replication of this macroeconomic phenomenon. The rapid increases of stock prices of AI-focused mega-cap companies demonstrate a massive accumulation of capital similar to these historical precedents.

2.2 Macroeconomic Regimes and Catalysts for Exuberance (2010–Present)

As technological innovation shapes the fundamental narrative for asset pricing, the macroeconomic environment determines the availability of capital necessary to fuel a speculative bubble. (Pástor and Veronesi, 2009) Over the past fifteen years, equity markets—specifically the Nasdaq index—have navigated four macroeconomic regimes, each characterized by unique liquidity constraints and investor behaviors.

2.2.1 The Quantitative Easing Era (2010–2019)

Following the Global Financial Crisis of 2008, the Federal Reserve and other major central banks implemented unprecedented monetary policies, at first, Zero Interest Rate Policies (ZIRP) and large-scale asset purchases known as Quantitative Easing (QE). This period of expansionary monetary policy essentially changed the mechanics of asset pricing. In a standard discounted cash flow (DCF) framework, lowering the risk-free rate increases the present value of future earnings.

As a result, capital was forced further out on the risk curve in a "search for yield". Growth equities, specifically in the technology sector, became the primary beneficiaries of this environment. The Nasdaq index performed an upward-trending path starting in 2010, showing a sustained period of capital accumulation in high-growth assets. This decade established a baseline where elevated valuation multiples became normalized, setting the stage for more extreme exuberance in the subsequent decade.

2.2.2 The Pandemic Shock and Hyper-Liquidity (2020–2021)

The COVID-19 pandemic in early 2020 caused a severe but short-lived market contraction. As a consequence, governments and central banks used fiscal and monetary stimulus on a scale not seen since the Second World War. This hyper-liquidity, mixed with structural economic shifts like global lock-downs and the transition to remote work, acted as a massive catalyst for the technology sector.

At the same time, the micro structure of the market matured. The expansion of zero-commission trading platforms and the distribution of fiscal stimulus checks equipped a new wave of retail investors. The retail influx accelerated herding behavior, where retail coordination drove relentless, compounding momentum in specific equities. Driven by both retail exuberance and institutional capital, the Nasdaq index experienced a sustained and significant increase in March 2020, ultimately peaking in December 2021.

2.2.3 The Inflationary Contraction

The pandemic era because of hyper-liquidity has eventually caused a high global inflation among the recent decades in 2022. Central banks ended ZIRP and started rate-hiking cycles in modern history.

When discount rates increased, the present value of future cash flows decreased, causing a severe repricing of growth equities. The Nasdaq Index entered a decline phase from the December 2021 peak to its lowest point in the cycle in November 2022. From an econometric perspective, this period would be critical for bubble testing, as it shows the "collapse" phase of a speculative regime.

2.2.4 The Generative AI Boom (2023 - Present)

Despite the increase in the interest rates, the technology sector recovered very sharply since December 2022. This recovery was caused by breakthroughs in Artificial Intelligence, and especially the mainstream commercialization of large language models and generative AI.

Unlike the meme stock frenzy of 2020, the AI boom is institutional, having large corporate capital expenditures into graphics processing units known as GPUs and high-speed parallel computation infrastructure. This concentration of capital was beneficial to mega-cap technology companies, known as "Magnificent Seven" (Alphabet, Amazon, Apple, Meta, Microsoft, Nvidia, and Tesla). By July 2024, these companies achieved a market capitalization that exceeds \$15.41 trillion together.

Nvidia, as the major and dominant producer of these GPU chips, has been affected the most in this market action. Nvidia's split-adjusted stock price increased by 571.14% from \$17.14 in December 2022 to \$115 in June 2024. This massive surge in the price of Nvidia demonstrates the dynamics of historical bubbles.

2.3 The Evolution of Econometric Bubble Detection

The detection of asset price bubbles relies on identifying explosive behavior in financial time series. However, the econometric methodology to precisely detect these regimes and behaviors has evolved rapidly, especially in mathematical challenges present in modeling periodically collapsing exuberance.

2.3.1 The Limitations of Standard Tests and the Evans Critique

Early trials to test for speculative bubbles depended on standard unit root and cointegration tests like Augmented Dickey-Fuller (ADF) test. The null hypothesis in a standard ADF framework, assumes the price of asset follows a random walk (a unit root process, $I(1)$), while the alternative hypothesis tests for explosive, non-stationary behavior.

Nonetheless, Evans (1991) showcased a severe limitation in this approach which is now known as the "Evans Critique". Evans proved that standard ADF tests lose significant power when they are applied to a time series that contains multiple bubbles that periodically emerge and collapse. Since the collapse phase of a bubble is violent downward trajectory, the standard ADF test when applied to the whole sample, absorbs this collapse as stationary behavior. Therefore, the test wrongly fails to reject the unit root null hypothesis which hides the presence of an explosive regime.

2.3.2 The First Breakthrough: PWY (2011)

To solve the Evans Critique issue, P. C. Phillips, Wu, and Yu (2011) introduced a shift in bubble detection methodology: the supremum Augmented Dickey-Fuller (SADF) test. Instead of testing the entire time series sample all at once, the PWY methodology does the ADF test recursively using a forward-expanding rolling window.

The PWY isolates the explosive segments before the collapse phase can neutralize this signal by calculating the ADF statistic for every sub-sample and taking the the supremum of these statistics. This methodology was highly effective in identification of isolated, singular bubble episodes, like the 1990s Dotcom bubble. Nevertheless, it still wasn't the most efficient when dealing with complex, multi-decade datasets that contained multiple overlapping bubble regimes.

2.3.3 The Real-Time Gold Standard: PSY (2015)

P. C. Phillips, Shi, and Yu (2015) generalized the PWY framework to address for multiple, periodically collapsing bubbles: the Generalized Supremum ADF (GSADF) and the Backward Supremum ADF (BSADF) tests.

The PSY methodology allows the starting point of the regression window to change recursively as well instead of expanding from a fixed starting point. The BSADF statistic is computed by running backward regressions from a fixed endpoint, while the GSADF calculates the supremum of the BSADF statistics across the whole sample. This architecture mathematically isolates each individual explosive regime which prevents a massive historical bubble from obscuring the detection of a subsequent smaller bubble.

2.3.4 The High-Frequency Data Trap and Time Aggregation

Even though the PSY framework is robust, there is still a critical challenge when utilizing it to modern financial data. High-frequency daily trading data involves extreme noise, persistent volatility clustering (heteroskedasticity), and price shocks.

Modern statistical packages must usually select a high number of lag length to account for the daily variance when applying autoregressive tests to daily data. Therefore, the extreme volatility and excessive lags can absorb the explosive autoregressive signal ($\hat{\theta}_n > 1$). The result is a false negative; the price is compounding explosively but the test considers the movement a cause of daily variance rather than a structural bubble regime.

Time aggregation is very necessary to maximize the statistical power of PSY methodology. Lowering the sample size to monthly or weekly time series strips

away the algorithmic volatility, leading to fundamental autoregressive compounding becoming mathematically legible to test.

2.4 Isolating Fundamentals from Price Action

A fundamental premise of asset pricing is that an increase in a stock's nominal price does not essentially mean a speculative bubble exists. If a stock's price increases by 500% over a two-year period, but the fundamental cash flows and revenues increase at an equal or greater magnitude, the valuation remains reasonable. Consequently, applying explosive autoregressive tests to raw nominal price can end up creating misleading results which means falsely identifying fundamental growth as speculative exuberance. To isolate pure, idiosyncratic speculation, econometricians must include fundamental metrics to strip away market beta.

2.4.1 The Role of Valuation Multiples

The literature traditionally uses financial ratios, scaling asset's market capitalization against its economic output to control for fundamental growth of the firm. The most common metric in bubble detection is the Price-to-Dividends or Price-to-Earnings ratio. When testing the ratio, researchers can understand if investors are paying an increasing irrational premium for the same unit of economic output.

However, the P/E ratio frequently fails when analyzing technological revolutions. Technology and biotechnology firms at very high growth rates, usually operate at a net loss for years since they reinvest capital into research and development. In such cases, the Price-to-Sales (P/S) ratio becomes the standard method for fundamental valuation. By using this metric under the PSY framework, the model tests if the movement of the stock price has decoupled from the revenues of the firm.

2.4.2 Factor Models and Idiosyncratic Exuberance

While valuation multiples do control for the fundamental growth of a firm, they do not account for the macroeconomic forces of the market. The entire Nasdaq index may experience an explosive behavior if the Federal Reserve injects large liquidity into the market. If a tech stock is tested during this period, the PSY test will probably flag a bubble, however, the cause behind this observation would remain unknown meaning whether the exuberance is a result of the market condition or the stock is experiencing its own idiosyncratic frenzy.

To make sure this doesn't happen, the literature suggests the decomposition of stock returns by using factor models, known as the Fama-French Three-Factor

Model (Fama and French, 1993). This framework considers the expected return of an asset is a function of: the overall market risk premium (Market Beta), the outperformance of small-cap companies compared to large-cap companies (SMB), and the outperformance of high book-to-market companies relative to low book-to-market companies. The decomposition can be mathematically expressed in the following regression:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_1(R_{M,t} - R_{f,t}) + \beta_2(SMB_t) + \beta_3(HML_t) + \varepsilon_{i,t} \quad (2.4.1)$$

Where $R_{i,t}$ is the individual stock return, $R_{f,t}$ is the risk-free rate, and $\varepsilon_{i,t}$ represents the residual, error term.

This residual error term ($\varepsilon_{i,t}$) is the critical component. When the accumulated residual is extracted from the regression, the Idiosyncratic Component (IC) is isolated. The IC represents the unexplained portion of the stock movement. Then, the PSY test is applied to this Idiosyncratic Component and if any explosive root is detected, it confirms the presence of an isolated speculative bubble.

2.5 Inference in Explosive Regimes

The more complex challenge comparing to bubble detection, would be the statistical inference on the severity of the bubble. When the asset shifts from a stationary or unit-root process into an explosive regime, traditional econometric limit theory breaks down. (P. C. Phillips, 2023)

2.5.1 The Challenge of Explosive Asymptotics

When an asset price follows a stationary ($I(0)$) or unit-root ($I(1)$) process, in time-series econometrics, Gaussian central limit theorems are used to construct standard errors and confidence intervals. However, when the price is experiencing a speculative bubble, the autoregressive coefficient exceeds unity ($\theta > 1$).

The time-series variance grows exponentially in explosive state. Therefore, it would cause the Gaussian asymptotic theory to be invalid. If the calculation of the standard confidence intervals for the autoregressive coefficient during a bubble is tried, the bands will be fundamentally flawed, which leads to the misinterpretations of asset's trajectory. The literature needs an alternative framework to measure the magnitude of the exuberance.

2.5.2 Near-Unit-Root Limit Theory (Phillips 2023)

To resolve this inferential problem, the near-unit-root framework by P. C. Phillips (2023) is used. Phillips modeled the autoregressive coefficient (θ_T) as a parameter

that drifts locally from unity as the sample size (T) increases rather than treating explosiveness as a fixed state.

The mathematical expression of the coefficient:

$$\theta_T = 1 + \frac{c}{T^\gamma} \quad (2.5.1)$$

Where c is a scale coefficient and γ is the localizing rate that represents how fast the process diverges from a standard random walk. The nature of the asset's regime can be categorized as following when the localizing rate is estimated:

- **Mildly Integrated (MIR):** When $\theta_T < 1$ and $0 < \gamma < 1$, the assets shows strong persistence but stays fundamentally mean-reverting.
- **Local-to-Unity (LUR):** When $\gamma \geq 1$, the process shows dynamics that are statistically indistinguishable from a standard unit root ($I(1)$), which represents a normal, non-speculative market environment.
- **Mildly Explosive (MER):** When $\theta_T > 1$ and $0 < \gamma < 1$, the price has entered a definitive speculative bubble. The smaller the γ , the more severe the explosive compounding.

2.5.3 Cauchy Asymptotics and Confidence Intervals

Phillips showed that when $\theta_T > 1$, the distribution of the test statistic does not converge to a normal distribution, but instead it converges to a standard Cauchy distribution.

Since the Cauchy distribution has heavy tails and undefined moments, the construction of confidence intervals for γ becomes sensitive. When an asset such as Nvidia or the Nasdaq index enters a speculative phase, applying Cauchy critical values generates asymmetric confidence intervals. If the upper bound of confidence interval is below 1, it can be stated that the asset is in a Mildly Explosive regime.

Chapter 3

Methodology

3.1 Data and Specification Strategy

A challenge in the detection of asset bubbles in econometrics is the understanding of rational price appreciation—caused by economic fundamentals or monetary policy—and irrational exuberance. Using a bubble detection algorithm to raw prices has the risk of identifying false positives, as a stock’s price may increase simply because of market trends.

To isolate pure idiosyncratic exuberance, we employ an econometric specification strategy that includes 3 distinct models:

3.1.1 Model 1: Price-to-Sales Ratio

The analysis starts with a baseline model, analyzing the Price-to-Sales (P/S) ratio of the asset. If explosive behavior is detected in this case, it shows that the stock’s price grew faster than its fundamental revenues. However, this is not an indication of a firm-specific speculative bubble, as the result could be the cause of systemic equity market trends or macroeconomic liquidity.

3.1.2 Model 2: Price-to-Index Ratio

To address the macroeconomic influence of model 1, the second specification uses Price-to-S&P 500 index. This controls for aggregate market beta. It tests whether the stock prices are increasing explosively relative to the broader economy. The S&P 500 is chosen instead of Nasdaq to prevent understating the outperformance since Nasdaq is heavily weighted toward the Magnificent Seven stocks.

3.1.3 Model 3: The Fama-French Idiosyncratic Component

To isolate firm-specific exuberance from equity market movements, this thesis follows the methodology used in the recent literature by Basele, Peter C B Phillips, and Shi (2025). In this case, stock returns are regressed against the Fama-French three factors (Excess Market Return, Small-Minus-Big, and High-Minus-Low). Standard equity risk premiums are stripped away by extracting and testing the residuals of this regression, leaving the idiosyncratic component of the stock.

$$R_{i,t} - R_{f,t} = \alpha + \beta_1(R_{M,t} - R_{f,t}) + \beta_2 SMB_t + \beta_3 HML_t + \varepsilon_{i,t}^{FF3} \quad (3.1.1)$$

Where:

- R_t is the return of the asset at time t .
- $R_{f,t}$ is the risk-free rate.
- $(R_{M,t} - R_{f,t})$ is the excess return of the broader market.
- SMB_t (Small-Minus-Big) controls for the size premium.
- HML_t (High-Minus-Low) controls for the value premium.
- $\varepsilon_{i,t}^{FF3}$ is the idiosyncratic residual.

3.1.4 Model 4: The Fama-French 5 Factor Idiosyncratic Component

The third specification extends the previous Fama-French three-factor approach by applying the Fama-French five-factor model. This model provides a stricter decomposition of stock returns because it controls not only for market, size, and value effects, but also for profitability and investment patterns. This is particularly relevant for the Magnificent Seven stocks, since these firms are usually characterized by high profitability, large intangible capital, and aggressive investment in technology, cloud infrastructure, artificial intelligence, and research and development.

The Fama-French five-factor model is written as:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_1(R_{M,t} - R_{f,t}) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 RMW_t + \beta_5 CMA_t + \varepsilon_{i,t}^{FF5} \quad (3.1.2)$$

Where:

- $R_{i,t}$ is the return of stock i at time t .
- $R_{f,t}$ is the risk-free rate.

- $(R_{M,t} - R_{f,t})$ is the excess market return.
- SMB_t is the size factor, measuring the return difference between small and large firms.
- HML_t is the value factor, measuring the return difference between high book-to-market and low book-to-market firms.
- RMW_t is the profitability factor, measuring the return difference between firms with robust and weak profitability.
- CMA_t is the investment factor, measuring the return difference between firms with conservative and aggressive investment behavior.
- $\varepsilon_{i,t}^{FF5}$ is the idiosyncratic residual component that remains unexplained by the five systematic risk factors.

After estimating the regression, the fitted values represent the part of each stock's return that can be explained by common risk factors. The residual term, $\varepsilon_{i,t}^{FF5}$, represents the unexplained component of the return. This residual is then accumulated over time to construct the Fama-French five-factor idiosyncratic component:

$$IC_{i,t}^{FF5} = \sum_{s=1}^t \varepsilon_{i,s}^{FF5} \quad (3.1.3)$$

The PSY test is then applied to this accumulated idiosyncratic component. If explosive behavior is detected in $IC_{i,t}^{FF5}$, this suggests that the stock contains a firm-specific speculative component that cannot be explained by market risk, size, value, profitability, or investment factors. Therefore, the five-factor specification provides the strictest test of idiosyncratic exuberance used in this thesis.

Compared with the three-factor model, the five-factor specification is especially useful for technology firms because it controls for profitability and investment, two characteristics that are central to the valuation of mega-cap growth companies. If a bubble remains statistically significant even after applying this specification, the evidence of firm-specific speculative behavior becomes stronger.

3.2 The GSADF Real-Time Detection Model

Once the idiosyncratic residuals are isolated, they are tested for explosive dynamics. Standard unit root tests, such as Augmented Dickey-Fuller (ADF) test, lose their accuracy in the presence of multiple market crashes. The volatility of an initial bubble collapse hides the statistical signature of subsequent bubbles in the sample.

To overcome this, the analysis relies on Generalized Supremum ADF (GSADF) methodology developed by P. C. Phillips, Shi, and Yu (2015). The test evaluates the null hypothesis of a random walk against the alternative hypothesis of mildly explosive autoregressive dynamics, defined by the following process:

$$y_t = \theta_T y_{t-1} + \epsilon_t \quad (3.2.1)$$

Where a coefficient of $\theta_T > 1$ indicates explosive, bubble-like behavior. To detect this without the interference of previous market crashes, the model estimates the ADF regression over a rolling window:

$$\Delta y_t = \hat{\alpha}_{r_1, r_2} + \hat{\beta}_{r_1, r_2} y_{t-1} + \sum_{i=1}^k \hat{\psi}_i \Delta y_{t-i} + \hat{\epsilon}_t \quad (3.2.2)$$

Where r_1 is the start of the sample window, r_2 is the end, and the window size is $r_w = r_2 - r_1$. Because the exact start and end dates of the speculative bubbles are unknown, the GSADF statistic calculates the maximum possible ADF t-statistics across all possible starting and ending points in the dataset:

$$GSADF(r_0) = \sup_{r_2 \in [r_0, 1], r_1 \in [0, r_2 - r_0]} \{ADF_{r_1}^{r_2}\} \quad (3.2.3)$$

The implementation of the GSADF test requires two key parameters: the minimum window size (r_0) and the lag length (k). Following the econometric literature, the minimum window size is determined based on the total sample size (T) using $r_0 = 0.01 + 1.8/\sqrt{T}$.

Sample Splitting and Training Period Contamination: A requirement of the P. C. Phillips, Shi, and Yu (2015) limit theory is the establishment of a minimum baseline window (τ_0). This period must consist of normal, non-explosive market behavior. However, the extreme volatility of the 2022 market crash may contaminate this baseline, resulting in a loss of statistical power of the test. To ensure robust inference, the sample is divided into pre-crash and post-crash periods.

3.3 Measuring Exuberance Severity (Phillips 2023)

While the GSADF test successfully detects the bubble, it does not measure its severity. In finite samples, the scale and rate coefficients of the explosive autoregressive roots are not separately identified. To solve it, this thesis uses the uniparameter localizing rate sequence (γ_n) developed by P. C. Phillips (2023).

The standard autoregressive model is re-parameterized to have both coefficients in a single metric:

$$\theta_n = 1 + \frac{1}{n^{\gamma_n}} \quad (3.3.1)$$

The parameter is then estimated using a non-linear rate estimator:

$$\hat{\gamma}_n = -\frac{\log |\hat{\theta}_n - 1|}{\log n} \quad (3.3.2)$$

If the asset enters an explosive regime ($\hat{\theta}_n > 1$), the limit distribution converges to a Cauchy distribution, free of nuisance parameters. This allows for the construction of 95% Confidence Intervals. Within the Mildly Explosive regime, smaller values of $\hat{\gamma}_n$ (moving further below 1) shows an exponentially faster rate of price divergence, providing a precise mathematical check for the severity of the asset's idiosyncratic exuberance.

3.4 Computational Implementation

To apply the PSY real-time bubble detection test on the metrics mentioned in this thesis, psymonitor package (Caspi, Peter C. B. Phillips, and Shi, 2024) in R is used. For the second period Generalized Augmented Dickey-Fuller tests in the Empirical section of Chapter 4, the exuber package (Vasilopoulos, Pavlidis, and Martínez-García, 2022) is utilized.

Chapter 4

Empirics

4.1 Speculative Bubbles

This chapter discusses the application of the theoretical frameworks mentioned in Chapter 3 with empirical market data to investigate the presence and also the severity of speculative bubbles in the Magnificent 7.

This analysis is divided into 2 periods. Period 1 starts from January 2010 through November 2021 and period 2 focuses on November 2021 onward.

The structure of the findings is as follows:

- **Period 1 Analysis:** P. C. Phillips, Shi, and Yu (2015) real-time bubble detection methodology is applied to P/S , P/I , The idiosyncratic components (IC) of Fama-French 3 and 5 factor models of the 7 stocks for the first period.
- **Period 2 Analysis:** The Generalized Supremum Augmented Dickey-Fuller (GSADF) test is applied to the second period of the stocks, and the test statistic are reported in the 4.1 table.
- **Bubble Severity Measurement:** The near-unit-root limit theory introduced by P. C. Phillips (2023) is utilized to calculate the localizing rate parameter to categorize each asset's regime as Mildly Integrated, Local-to-Unity, or Mildly Explosive.

4.2 Period 1

In this period, the PSY real-time detection procedure is done to capture speculative bubbles and identify the start and the end of the speculative phase. The analysis demonstrates this process over 4 metrics, each imposing progressively stricter controls for fundamental and systemic elements. The Price-to-Sales ratio

acts as a fundamental baseline. The Price-to-S&P 500 Index ratio controls for aggregate market beta. The Fama-French three factor and five factor idiosyncratic components account for the systematic risk related to size, value, profitability, and investment. If a bubble survives all of these four specifications, it can be an evidence of genuinely idiosyncratic speculative exuberance.

4.2.1 Price-to-Sales Ratio

In this part, the PSY test is applied to the Price-to-Sales ratio of each Magnificent Seven stock. As mentioned in Section 2.4.1, the P/S ratio is an appropriate fundamental control for high-growth technology companies, some of which operated at low or negative earnings during some periods of the sample due to capital reinvestment. Explosive behavior in this ratio shows that the prices have grown faster than firm revenues.

Among the results in Figure 4.1, Alphabet, Amazon, and Meta do not produce clear speculative signals under this specification, which suggests that their price appreciation in this period was followed by growth in their revenues. However, other stocks show signals of speculative bubbles, with the most prominent one being Nvidia which shows multiple speculative episodes, with one lasting almost a year and two of them being right at the peaks of P/S in 2021 and 2022. Tesla and Apple had only one month of statically significant speculative episodes meaning starting and ending in the same month. Moreover, Microsoft shows evidence of shorter-lived speculative episodes under the P/S specification.

It is important to note that the P/S ratio does not isolate firm-specific exuberance from macroeconomic or sector-wide forces. This motivates the transition to market and factor specifications.

4.2.2 Price-to-S&P 500 Index Ratio

Figure 4.2 addresses the PSY test results for the ratio of adjusted stock prices to the S&P 500 index. By utilizing this metric, we can control for aggregate market beta which can test whether the individual stock prices are increasing explosively relative to the broad economy. This approach is consistent with the Price-to-Index methodology employed by P. C. Phillips, Shi, and Yu (2015), with the distinction that the S&P 500 is used here rather than Nasdaq. The reason behind this is since Nasdaq is heavily weighted toward technology sector and a massive proportion of it is composed by Magnificent Seven stocks, using it as a benchmark would risk understating the degree of outperformance. Therefore, S&P 500 is more appropriate for the purpose of detecting exuberance from the broader economy.

Under this specification, the density of detected speculative episodes increases substantially. By accounting for aggregate market, the test now identifies stocks whose price have diverged from the broader economy. Among them, the most prominent one is Microsoft which shows a large continuous speculative episode starting from mid 2018 almost through the end of 2022. Nvidia exhibits multiple distinct speculative episodes across the period, with the two shaded areas in 2021 and onward suggesting the significant increase in demand of AI data centers. Moreover, Amazon, and Apple demonstrate evidence of explosive dynamics especially at the peaks of their P/I , which interestingly are followed by periods of decline. On the other hand, Tesla, Alphabet, and Meta show small evidences of speculative episodes throughout their sample suggesting weaker signals of bubble dynamics.

4.2.3 Fama-French 3 Factor Idiosyncratic Components

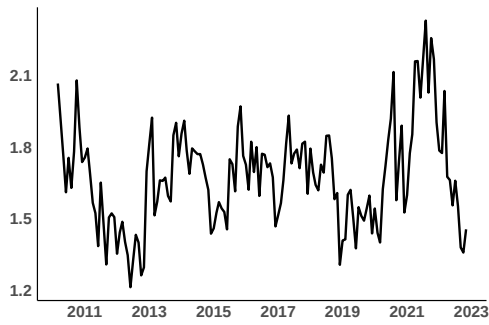
Figure 4.3 addresses PSY test results for the idiosyncratic components which were extracted from the Fama-French three-factor model, as mentioned in Section 3.1. The three factor model accounts for excess market return, the SMB size premium, and the HML value premium from each stock's return series. The accumulated residuals represent the portion of the price which is not represented by market movement and cross-sectional risk factors. Explosive behavior in this component suggests evidence of speculative activity which is firm-specific.

Based on Figure 4.3, the results are mostly consistent with earlier specifications, but providing more precise identification of exuberance. Microsoft still shows statistically significant explosive behavior, however, more specific phases comparing to P/I specification. The persistence of this speculative phase suggests strong evidence that Microsoft's premium cannot be attributed to fundamental revenue growth, aggregate market appreciation, or systematic risk factors. Nvidia, continues producing the same large bubble episode from mid 2016 toward the end of 2018, maintaining a strong presence across all three specifications mentioned so far. On the other hand, it can be observed that the visible speculative episodes covering 2021 until 2022, are barely present anymore disappearing when accounting for systematic risk factors. Meta contains more short-period speculative episodes comparing to P/I specification. However Apple behaves differently by showing significantly less bubble periods comparing to the earlier specification. Tesla shows a stronger speculative presence in mid 2013 compared to earlier applications. Alphabet and Amazon show no evidence of idiosyncratic exuberance under this specification, recommending a more anchored valuation to systematic factors throughout this sample period.

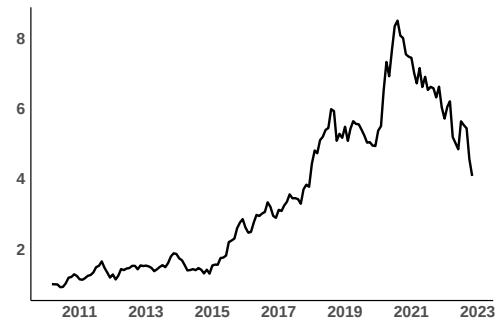
4.2.4 Fama-French 5 Factor Idiosyncratic Components

Figure 4.4 expands the IC analysis to the five factor model, containing Robust-Minus-Weak profitability factor and the Conservative-Minus-Aggressive investment factor. This specification represents the most comprehensive control for systematic influences and constitutes an extension beyond the methodology of Basele, Peter C B Phillips, and Shi (2025), which relies exclusively on the three-factor model. The two additional factors for this specification are a good fit especially for Magnificent 7 stocks due to their mainly high profitability and aggressive investment.

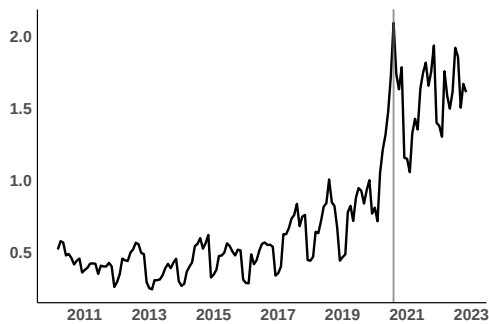
The results in Figure 4.4 provide some small differences compared to the three factor model. Microsoft tends to behave the same way but with more detailed bubble periods compared to the earlier specification. The continuous statistically significant speculative episode seems to be divided more, therefore suggesting a higher accuracy of the speculative dates. Nvidia on the other hand, exhibits the same massive episode and also a small difference in the 2022 phase which has gotten larger than it was in the earlier application. Moreover, Tesla seems to be showing the same behavior, evident in the 2022 phase being larger than in the previous specification. On the other hand, a decrease in the number of speculative episodes can be seen in Apple and Meta. Alphabet and Amazon remain the same with showing no evidence of statistically significant idiosyncratic exuberance.



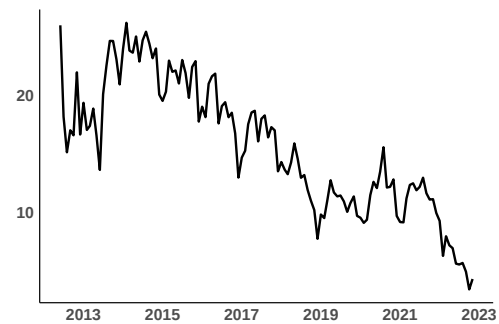
(a) Alphabet



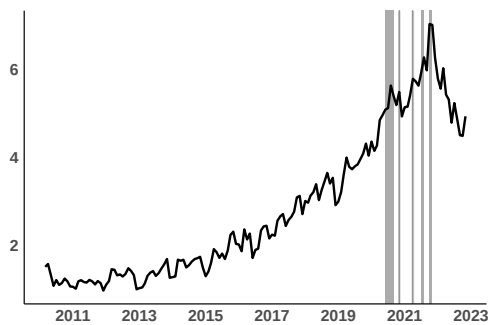
(b) Amazon



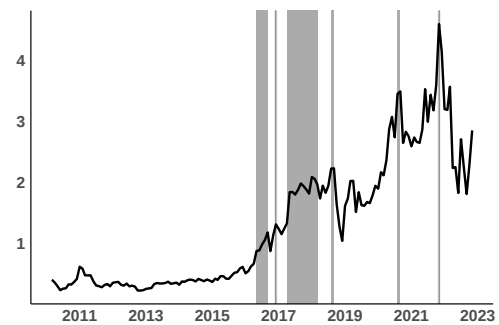
(c) Apple



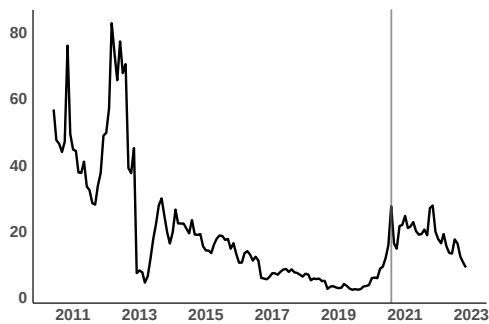
(d) Meta



(e) Microsoft



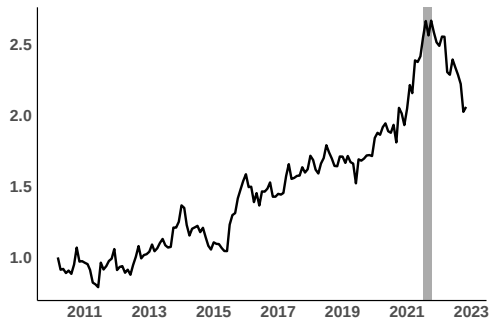
(f) Nvidia



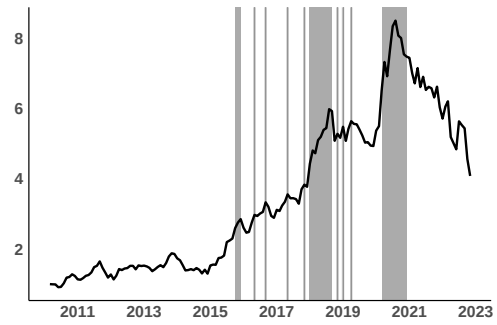
(g) Tesla

Note: This figure displays results of the PSY test applied to the ratio of the (adjusted) Mag-7 stock prices to sales. The shaded areas show periods where the PSY statistic exceeds 90% bootstrap critical value (2000 replications) over Period 1 sample. Lag length selection is based on Bayesian Information Criterion with a max lag of 6.

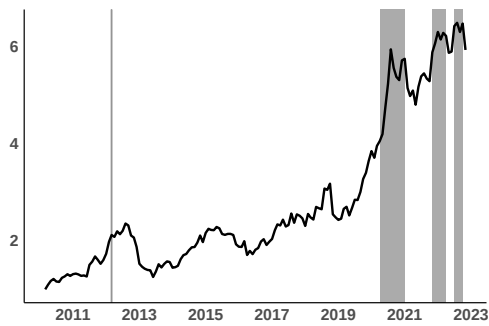
22
Figure 4.1: PSY Test on the Price to Sales Ratio



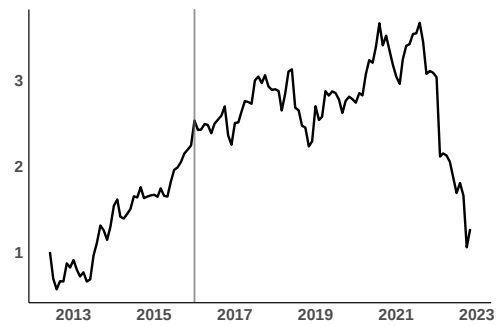
(a) Alphabet



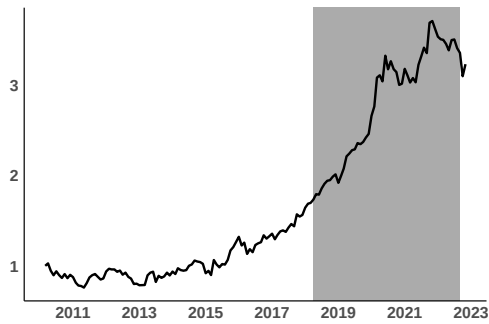
(b) Amazon



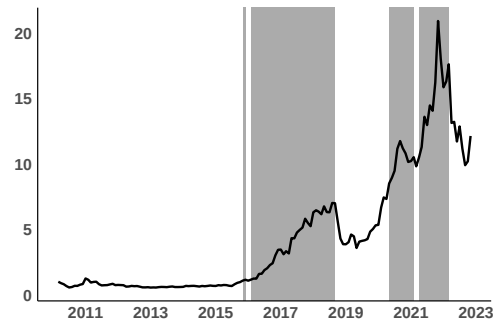
(c) Apple



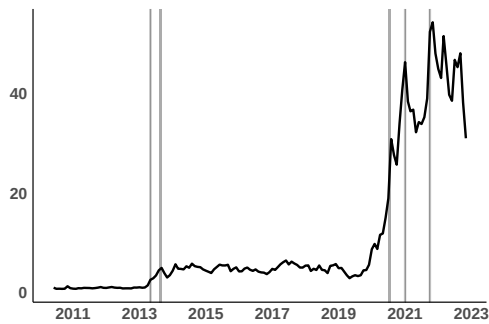
(d) Meta



(e) Microsoft



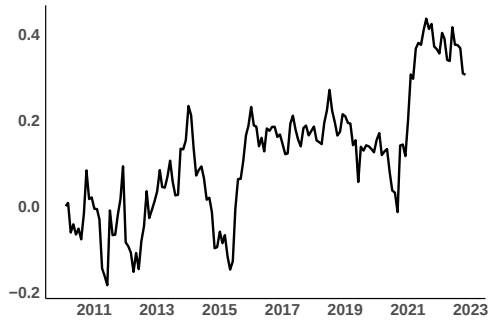
(f) Nvidia



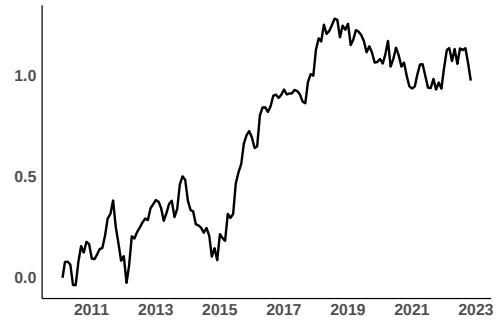
(g) Tesla

Note: This figure displays results of the PSY test applied to the ratio of the (adjusted) Mag-7 stock prices to S&P 500 Index. The shaded areas show periods where the PSY statistic exceeds 90% bootstrap critical value (2000 replications) over Period 1 sample. Lag length selection is based on Bayesian Information Criterion with a max lag of 6.

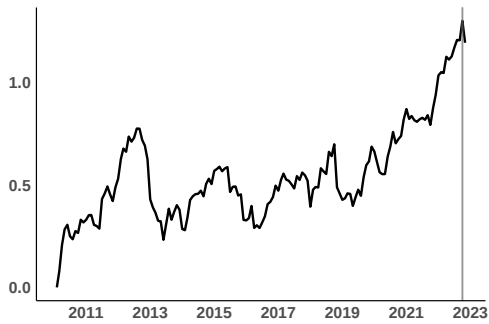
Figure 4.2: PSY Test on the Price to S&P500 Ratio



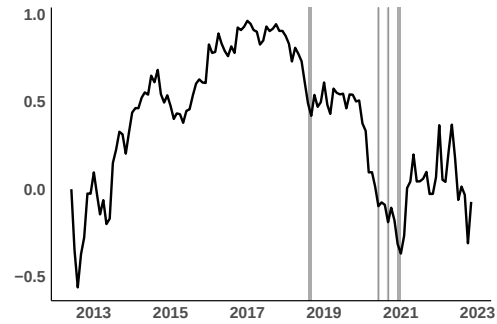
(a) Alphabet



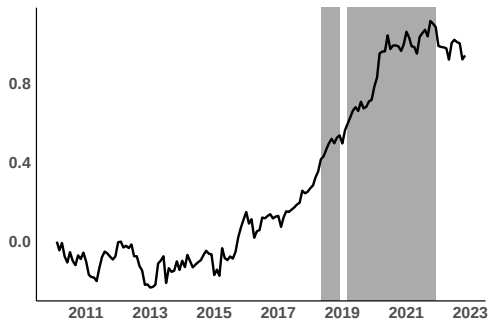
(b) Amazon



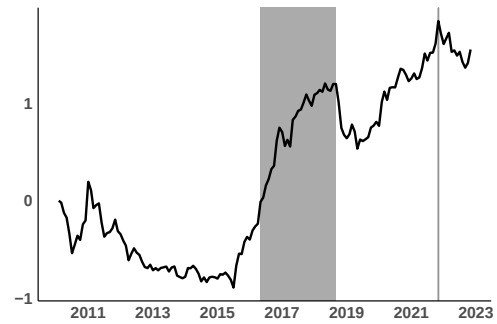
(c) Apple



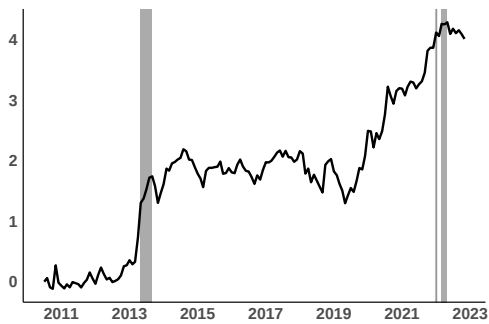
(d) Meta



(e) Microsoft



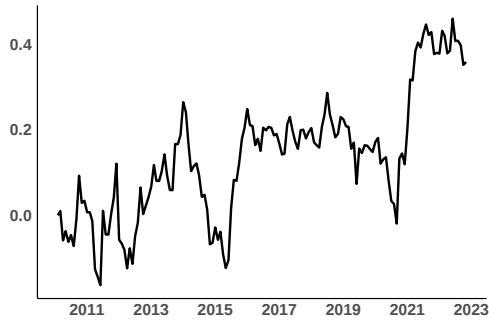
(f) Nvidia



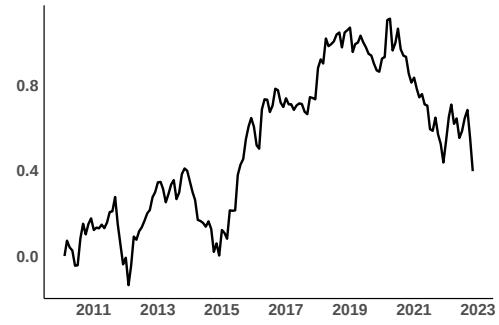
(g) Tesla

Note: This figure displays results of the PSY test applied to the idiosyncratic components (IC) of the Mag-7 stock based on Fama-French 3 factor model. The shaded areas show periods where the PSY statistic exceeds 90% bootstrap critical value (2000 replications) over Period 1 sample. Lag length selection is based on Bayesian Information Criterion with a max lag of 6.

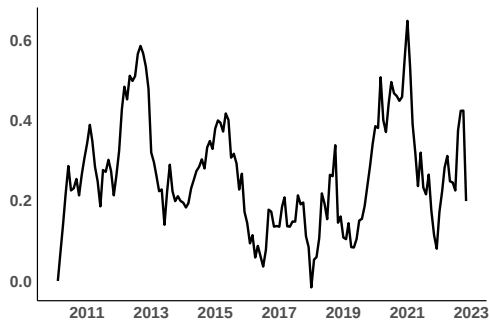
Figure 4.3: PSY Test on the IC Components (Fama-French 3 factor)



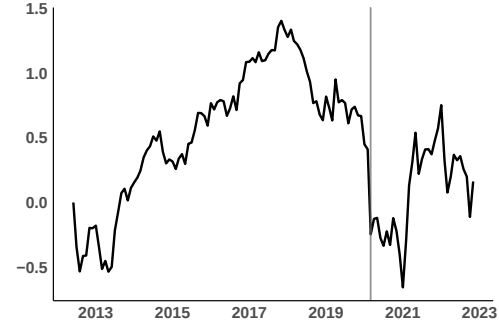
(a) Alphabet



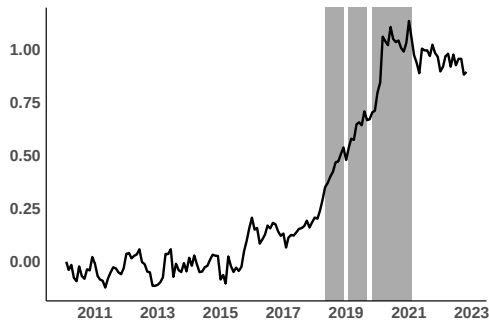
(b) Amazon



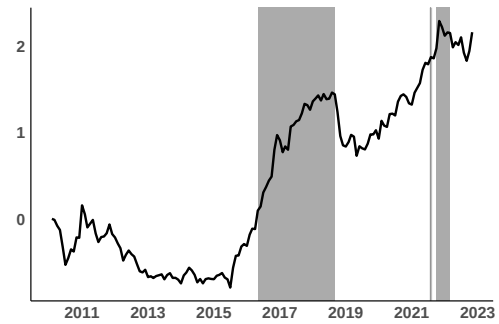
(c) Apple



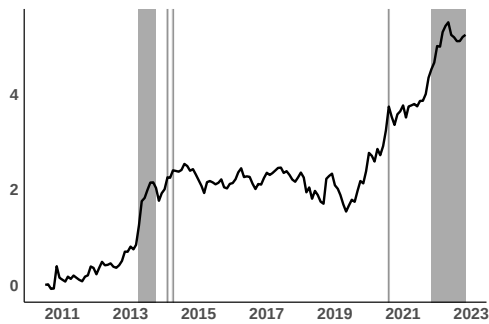
(d) Meta



(e) Microsoft



(f) Nvidia



(g) Tesla

Note: This figure displays results of the PSY test applied to the idiosyncratic components (IC) of the Mag-7 stock based on Fama-French 5 factor model. The shaded areas show periods where the PSY statistic exceeds 90% bootstrap critical value (2000 replications) over Period 1 sample. Lag length selection is based on Bayesian Information Criterion with a max lag of 6.

Figure 4.4: PSY Test on the IC Components (Fama-French 5 factor)

4.2.5 Summary of Period 1 Findings

Taken together, the four specifications show a consistent picture of speculative activity across the Magnificent Seven during 2010-2022. Microsoft and Nvidia are the two stocks for which bubble episodes are detected more persistently across all specifications. This demonstrates the strongest evidence of idiosyncratic speculative exuberance within this framework. Meta and Alphabet exhibited limited evidence of idiosyncratic exuberance regardless of the specification. Others were variable dependent on the specification utilized.

4.3 Period 2

The second sample periods starts from December 2022 and ends at June 2026. As discussed in Section 3.2, the short duration of this period makes the full PSY real-time methodology inappropriate for bubble date-stamping, since the procedure requires a sufficient non-explosive training window prior to each detected episode. Instead, the Generalized Supremum Augmented Dickey-Fuller statistic is used across all four specifications to test the null hypothesis of no explosive dynamics within the period. Table 4.1 reports the results for each stock under each methodology.

The main finding of this period is that across all four specifications and all seven stocks, none of the GSADF test statistics are statistically significant. Under the P/S specification, Alphabet has the highest statistic of 1.86, followed by Nvidia at 1.21 under the P/I specification, but neither value exceeds the critical values required for rejection of the unit root null. The remaining statistics are much lower, with several of them having negative values – such as Amazon and Apple under P/S ratio, with values of -1.02 and -1.21 respectively – which indicates deviation of price dynamics in the stationary rather than the explosive direction relative to their fundamentals. The FF3 and FF5 IC specifications produce similar uniform statistics, highest being 0.84 and 0.83 for Microsoft under both factor models, which are well below the critical thresholds. Taken together, these results provide no econometric evidence of speculative bubbles among the Magnificent Seven from December 2022 to June 2026, under any of the four specifications utilized.

However, these findings differ from the reference study, which reports significant GSADF statistics for several Magnificent Seven stocks during the second period. A likely explanation is the difference in the sample length. The article’s second period ends in January 2025 and includes only 26 observations, capturing the sharp initial phase of the AI-related rally. On the other hand, this thesis extends the sample to June 2026, increasing the sample to 43 observations. The additional observations may smooth the early explosive signal and incorporate later market

adjustments in valuations. Consequently, the evidence of exuberance becomes weaker when the post-boom period is analyzed over a longer time span. Also, it's worthy to mention that different minimum windows – such as 12, and 24 – have been tested to check the robustness, and yet still the results stayed the same in terms of statistical significance.

Overall, The period 2 results suggest that the AI-related rally after December 2022 does not constitute a statistically confirmed speculative bubbles under the GSADF framework used in this thesis. Although several stocks, especially Nvidia, Alphabet, and Microsoft, experienced substantial price increases, the econometric results do not reject the null hypothesis of no explosive dynamics. Therefore, in this 43 observation sample, the appreciation of the Magnificent Seven appears strong but non-explosive, rather than a formally detectable speculative bubble.

Table 4.1: GSADF Test Statistics for Mag-7 Stocks (Period 2)

Methodology	Alphabet	Amazon	Apple	Meta	Microsoft	Nvidia	Tesla
Price/Sales (P/S)	1.86	-1.02	-1.21	-0.67	-0.09	-1.13	-0.09
Price/Index (P/I)	1.31	-0.36	-1.01	-0.27	0.03	1.21	-0.73
FF3 IC	0.35	0.025	-0.18	-0.02	0.84	-0.49	0.42
FF5 IC	0.32	-0.95	0.12	-0.17	0.83	-0.45	-0.45

Note: This table presents the Generalized Sup ADF (GSADF) test statistics on Period 2 of all specification of the Magnificent Seven stocks (43 observations) with 90% critical values, minimum window size set to 18, and 2000 replications with exuber package in R (Vasilopoulos, Pavlidis, and Martínez-García, 2022).

4.4 Regime Behavior

The previous sections focused on the detection of speculative episodes in the first and second period samples of Magnificent Seven stocks, however, identifying a whether a bubble exists is only the first part of the empirical analysis. The second part focuses on an equally important concept, which is the intensity or the explosiveness of the regimes meaning whether an asset shows persistent or explosive behavior.

For this purpose, this section applies the near-unit-root inference framework of P. C. Phillips (2023) on full samples of Period 1 and 2. The objective is more than just classifying each stock as having or not having a bubble. It's to measure the rate at which the autoregressive process moves away from unity. This is calculated through estimating the autoregressive coefficient $\hat{\theta}_n$, the localizing rate parameter $\hat{\gamma}_n$, and the confidence interval for γ . The interpretation follows the logic introduced in Chapter 3. When $\hat{\theta}_n > 1$, the process is on the explosive side of unity. If the confidence interval for γ is entirely below 1, the process is

classified as Mildly Integrated. Finally, when the confidence interval includes one, the evidence is more consistent with Local-to-Unity behavior, meaning that the process remains very persistent but statistically close to a unit root.

Tables 4.2 to 4.5 report the severity results for the four specifications used throughout this chapter: Price-to-Sales, Price-to-Index, the Fama-French 3 factor and 5 factor idiosyncratic components. The results are presented for both periods.

4.4.1 Price-to-Sales Ratio Severity Measurement

Starting with the Price-to-Sales methodology in table 4.2, the results show that most stocks do not show a statistically confirmed Mildly Explosive regime. In period 1, Microsoft is the only stock with an autoregressive coefficient above unity, with $\hat{\theta}_n = 1.0039$, but its confidence interval for γ , equal to $(0.0000, 2.5003)$, includes one. Therefore, this cannot be classified as a robust Mildly Explosive process. Tesla records the lowest $\hat{\gamma}_n$ in Period 1, equal to 0.6162, with a confidence interval of $(0.4382, 0.7941)$, which is entirely below one. However, since its autoregressive coefficient is below unity, $\hat{\theta}_n = 0.9542$, this result indicates Mildly Integrated behavior rather than Mildly Explosive behavior. In other terms, Tesla's P/S ratio in Period 1 appears highly persistent but mean-reverting, not explosively diverging from fundamentals.

In Period 2, the P/S results move slightly toward the explosive side for several stocks. Alphabet, Amazon, Apple, Meta, and Tesla all have $\hat{\theta}_n > 1$. However, their confidence intervals are very wide and include 1. For example, Alphabet records $\hat{\theta}_n = 1.0173$ and $\hat{\gamma}_n = 1.0794$, but the confidence interval is $(0.0000, 2.6883)$. Tesla also has $\hat{\theta}_n = 1.0134$, but its confidence interval is $(0.0000, 3.0820)$ is too wide to confirm a Mildly Explosive process. Therefore, under the P/S specification, period 2 shows some point estimates on the explosive side, but the uncertainty around γ prevents a strong severity classification.

Table 4.2: Autoregressive Coefficient Estimates, Localizing Rate Estimates, and 95% Confidence Intervals for Magnificent 7 Stocks P/S specification

Stock	Period	$\hat{\theta}_n$	$\hat{\gamma}_n$	$CI_{\hat{\gamma}}$
Alphabet	1	0.9936	1.0045	(0.5362, 1.4727)
	2	1.0173	1.0794	(0.0000, 2.6883)
Amazon	1	0.9899	0.9147	(0.5410, 1.2884)
	2	1.0068	1.3279	(0.0000, 3.8390)
Apple	1	0.997	1.1593	(0.4685, 1.8500)
	2	1.0077	1.2943	(0.0000, 3.7097)
Meta	1	0.983	0.8442	(0.5134, 1.1749)
	2	1.003	1.5427	(0.0000, 4.4916)
Microsoft	1	1.0039	1.1051	(0.0000, 2.5003)
	2	0.9977	1.6132	(0.0000, 3.5729)
Nvidia	1	0.9959	1.0941	(0.5076, 1.6805)
	2	0.9842	1.109	(0.3460, 1.8721)
Tesla	1	0.9542	0.6162	(0.4382, 0.7941)
	2	1.0134	1.1531	(0.0000, 3.0820)

4.4.2 Price-to-Index Ratio Severity Measurement

The Price-to-Index methodology in Table 4.3 gives evidence. In Period 1, Alphabet, Amazon, Apple, Microsoft, and Nvidia have autoregressive coefficients slightly above one. Among these, Apple and Microsoft have $\hat{\gamma}_n$ values below one, equal to 0.9309 and 0.9749 respectively. These estimates suggest stronger persistence and possible explosive pressure. However, their confidence intervals both include one, meaning that the evidence is not strong enough to classify them as Mildly Explosive. Nvidia, despite its importance in the technology and AI, records $\hat{\theta}_n = 1.0035$ and $\hat{\gamma}_n = 1.1282$, with a confidence interval of (0.0000, 2.6161). This again supports Local-to-Unity behavior rather than a statistically confirmed Mildly Explosive regime.

In Period 2, the P/I methodology shows Nvidia as the strongest one for explosive behavior. Nvidia has the largest autoregressive coefficient in this specification, with $\hat{\theta}_n = 1.0193$, and the lowest $\hat{\gamma}_n$ among the explosive-side stocks, equal to 1.0557. This indicates that Nvidia's price relative to the index was the closest to an explosive pattern in the second period. Nevertheless, the confidence interval (0.0000, 2.5675) includes one, so the result remains statistically consistent

with Local-to-Unity category. Alphabet also shows an explosive coefficient, with $\hat{\theta}_n = 1.0136$, but its confidence interval is similarly wide containing 1. Overall, the P/I results indicate that the second period, although having strong relative performance for some stocks, especially Nvidia, still is not a Mildly Explosive regime.

Table 4.3: Autoregressive Coefficient Estimates, Localizing Rate Estimates, and 95% Confidence Intervals for Magnificent 7 Stocks P/I specification

Stock	Period	$\hat{\theta}_n$	$\hat{\gamma}_n$	$CI_{\hat{\gamma}}$
Alphabet	1	1.0032	1.1454	(0.0000, 2.6994)
	2	1.0136	1.1428	(0.0000, 3.0213)
Amazon	1	1.0007	1.4387	(0.0000, 3.6896)
	2	1.0034	1.5101	(0.0000, 4.4096)
Apple	1	1.0093	0.9309	(0.3164, 1.5453)
	2	1.0025	1.589	(0.0000, 4.5996)
Meta	1	0.9977	1.2578	(0.3600, 2.1557)
	2	1.0043	1.4517	(0.0000, 4.2491)
Microsoft	1	1.0075	0.9749	(0.1633, 1.7864)
	2	0.9957	1.4477	(0.0123, 2.8831)
Nvidia	1	1.0035	1.1282	(0.0000, 2.6161)
	2	1.0193	1.0557	(0.0000, 2.5675)
Tesla	1	0.9987	1.3299	(0.2688, 2.3909)
	2	0.9991	1.885	(0.0000, 5.1386)

4.4.3 FF3 IC Ratio Severity Measurement

The Fama-French three-factor idiosyncratic component results in Table 4.4 provide a stricter test because systematic market, size, and value effects are removed. In Period 1, Apple, Microsoft, Nvidia, and Tesla have autoregressive coefficients above unity. Tesla and Apple have relatively low $\hat{\gamma}_n$ estimates, equal to 0.9341 and 0.9412, while Microsoft is also close to the Mildly Explosive boundary with $\hat{\gamma}_n = 0.9867$. However, in all three cases, the confidence intervals include one. This means that the point estimates suggest possible idiosyncratic exuberance, but the confidence intervals do not provide enough statistical precision to classify these stocks as Mildly Explosive.

An important result in Table 4.4 is Alphabet in Period 1. Alphabet has $\hat{\theta}_n = 0.9751$, $\hat{\gamma}_n = 0.7339$, and a confidence interval of (0.4971, 0.9707), which lies

entirely below one. Since the autoregressive coefficient is below unity, this result indicates Mildly Integrated behavior. Therefore, Alphabet’s idiosyncratic component under the FF3 methodology appears strongly persistent but mean-reverting. This is not evidence of a speculative bubble; instead, it suggests a sub-unit-root dynamic that remains below the explosive boundary.

In Period 2, the FF3 idiosyncratic component results again do not support a clear Mildly Explosive classification. Alphabet has the highest explosive-side autoregressive coefficient, $\hat{\theta}_n = 1.0307$, and a relatively low $\hat{\gamma}_n = 0.9444$. This is one of the strongest point-estimate signals in the severity analysis. However, the confidence interval (0.0000, 1.9662) includes one, meaning that Local-to-Unity behavior cannot be rejected. Nvidia and Meta also have autoregressive coefficients above unity, but their $\hat{\gamma}_n$ estimates are above one and their intervals are wide. Therefore, after controlling for the Fama-French three factors, the post-2022 idiosyncratic components do not show robust explosive severity.

Table 4.4: Autoregressive Coefficient Estimates, Localizing Rate Estimates, and 95% Confidence Intervals for Magnificent 7 Stocks Fama-French 3 factor Idiosyncratic Components (IC) specification

Stock	Period	$\hat{\theta}_n$	$\hat{\gamma}_n$	$CI_{\hat{\gamma}}$
Alphabet	1	0.9751	0.7339	(0.4971, 0.9707)
	2	1.0307	0.9444	(0.0000, 1.9662)
Amazon	1	1.0022	1.2123	(0.0000, 2.9931)
	2	1.0072	1.3361	(0.0000, 3.9017)
Apple	1	1.0088	0.9412	(0.2827, 1.5997)
	2	0.9919	1.3066	(0.1965, 2.4167)
Meta	1	0.9778	0.7883	(0.4993, 1.0773)
	2	1.0108	1.2283	(0.0000, 3.4585)
Microsoft	1	1.007	0.9867	(0.1219, 1.8514)
	2	0.9942	1.3956	(0.0876, 2.7035)
Nvidia	1	1.0039	1.1022	(0.0000, 2.4840)
	2	1.0109	1.2246	(0.0000, 3.4417)
Tesla	1	1.0095	0.9341	(0.2994, 1.5688)
	2	1.0051	1.4319	(0.0000, 4.2269)

4.4.4 FF5 IC Ratio Severity Measurement

The Fama-French five-factor idiosyncratic component results in Table 4.5 confirm the same general conclusion. In Period 1, Microsoft, Nvidia, and Tesla have autoregressive coefficients above unity. Nvidia and Tesla show the strongest point-estimate evidence of possible explosive behavior, with $\hat{\gamma}_n = 0.9323$ for Nvidia and $\hat{\gamma}_n = 0.8797$ for Tesla. These values are below one, suggesting faster divergence from the unit-root benchmark. However, both confidence intervals include one. Therefore, even under the FF5 methodology, the evidence is not sufficient to classify either stock as Mildly Explosive.

In Period 2, Alphabet, Amazon, Meta, Nvidia, and Tesla all have autoregressive coefficients slightly above unity under the FF5 specification. Alphabet has $\hat{\theta}_n = 1.0269$ and $\hat{\gamma}_n = 0.9804$, making it the closest to a Mildly Explosive point estimate in this specification. Meta and Nvidia also show explosive-side coefficients, equal to 1.0140 and 1.0094 respectively. However, all corresponding confidence intervals include one. Tesla has an autoregressive coefficient very close to unity, $\hat{\theta}_n = 1.0006$, and a high $\hat{\gamma}_n = 2.0270$, which is strongly consistent with near-unit-root rather than explosive behavior.

Table 4.5: Autoregressive Coefficient Estimates, Localizing Rate Estimates, and 95% Confidence Intervals for Magnificent 7 Stocks Fama-French 5 factor Idiosyncratic Components (IC) specification

Stock	Period	$\hat{\theta}_n$	$\hat{\gamma}_n$	$CI_{\hat{\gamma}}$
Alphabet	1	0.9813	0.7907	(0.5175, 1.0639)
	2	1.0269	0.9804	(0.0000, 2.1657)
Amazon	1	0.9960	1.0997	(0.5055, 1.6940)
	2	1.0114	1.2140	(0.0000, 3.3933)
Apple	1	0.9807	0.7847	(0.5156, 1.0538)
	2	0.9247	0.7009	(0.3377, 1.0642)
Meta	1	0.9752	0.7660	(0.4922, 1.0399)
	2	1.0140	1.1573	(0.0000, 3.1206)
Microsoft	1	1.0056	1.0292	(0.0000, 2.0901)
	2	0.9873	1.1843	(0.2984, 2.0701)
Nvidia	1	1.0092	0.9323	(0.3129, 1.5516)
	2	1.0094	1.2648	(0.0000, 3.6180)
Tesla	1	1.0123	0.8797	(0.4674, 1.2921)
	2	1.0006	2.0270	(0.0000, 5.3735)

4.4.5 Summary of Bubble Severity

Taken together, the severity results show that the Magnificent Seven stocks are generally characterized by high persistence rather than statistically confirmed explosive severity. Several stocks produce point estimates that are economically interesting. Nvidia appears as an important candidate under the P/I methodology in Period 2. Alphabet shows a strong explosive-side coefficient under the FF3 and FF5 specifications in Period 2. Tesla shows the lowest $\hat{\gamma}_n$ under P/S in Period 1, although its coefficient is below unity and therefore indicates Mildly Integrated behavior. However, once the confidence intervals are considered, almost all cases include $\gamma = 1$, meaning that Local-to-Unity dynamics cannot be rejected.

This finding is important because it refines the interpretation of the bubble detection results. The strong price increases observed in the Magnificent Seven, especially during the AI-related recovery, may visually resemble explosive market behavior. However, the near-unit-root severity analysis suggests that these dynamics are not statistically far enough from the unit-root boundary to be classified as robust Mildly Explosive processes. In other words, the stocks show persistence and strong growth, but the estimated explosive component is not sufficiently precise to confirm severe speculative dynamics.

Compared with the reference study, the results of this thesis are more conservative. (Basele, Peter C B Phillips, and Shi, 2025) find clearer evidence of speculative behavior and report Mildly Explosive dynamics for some stocks in the earlier period, especially Nvidia and Tesla. In this thesis, the wider sample extension to June 2026 and the use of four alternative methodologies lead to weaker severity classifications. The broader sample appears to absorb part of the early AI-boom acceleration and produces confidence intervals that generally include the Local-to-Unity case.

The overall conclusion from the bubble severity analysis is therefore that the Magnificent Seven stocks do not display robust Mildly Explosive behavior under the specifications used in this thesis. Some individual estimates suggest potential exuberance, particularly for Nvidia, Alphabet, Apple, Microsoft, and Tesla in specific models and periods. Nevertheless, the confidence intervals show that these cases remain statistically compatible with Local-to-Unity behavior. Therefore, the empirical evidence supports the presence of strong persistence and valuation pressure, but not a severe and statistically confirmed speculative bubble across the Magnificent Seven.

Conclusion

This thesis investigated whether the rapid increase in the valuation of the Magnificent Seven stocks reflects speculative bubble behavior or whether it can be interpreted as strong but non-explosive growth. The motivation for this research comes from the recent concentration of capital in the technology sector, especially after the commercialization of generative artificial intelligence. Since the Magnificent Seven firms represent a large share of modern equity market performance, understanding whether their price dynamics contain speculative components is important for both empirical finance and risk management.

The main challenge in this research was that a simple increase in stock prices cannot be automatically interpreted as a bubble. A technology company may experience strong price appreciation because of real improvements in sales, earnings expectations, market dominance, or exposure to broader macroeconomic and sectoral trends. For this reason, this thesis avoided relying only on nominal stock prices. Instead, the empirical analysis used four different specifications: the Price-to-Sales ratio, the Price-to-Index ratio, the Fama-French three-factor idiosyncratic component, and the Fama-French five-factor idiosyncratic component. This structure allowed the analysis to move gradually from a fundamental valuation control to stricter controls for systematic risk factors.

The empirical analysis was divided into two periods. Period 1 covered the pre-crash and pandemic-era phase, from January 2010 to November 2021. In this period, the P. C. Phillips, Shi, and Yu (2015) real-time bubble detection methodology was applied to identify the origination and termination of speculative episodes. The results show that speculative signals were present among the Magnificent Seven, but they were not equally distributed across firms. Microsoft and Nvidia showed the most persistent evidence of bubble episodes across the different specifications. This is an important result because their speculative signals remained visible even after controlling for broader market movements and systematic risk factors. Therefore, these two stocks provide the strongest evidence of idiosyncratic speculative exuberance in the first period.

The results for the other stocks were more mixed. Apple, Tesla, and Meta showed some evidence of speculative episodes, but these results depended more

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strongly on the specification used. Alphabet and Amazon showed weaker evidence of idiosyncratic exuberance, especially after the analysis moved from simple valuation ratios to factor-based residual components. This suggests that, for some stocks, apparent price exuberance may be largely explained by market-wide or fundamental factors rather than by firm-specific speculative dynamics.

Period 2 covered the post-2022 AI-related recovery, from December 2022 to June 2026. Because this period was shorter and because the initial observations may already belong to the AI-driven recovery phase, the full PSY date-stamping method was not used. Instead, the GSADF test was applied to each stock and each specification. The results of this period are more conservative than the findings of the reference study. Across all four methodologies and all seven stocks, none of the GSADF statistics were statistically significant. Therefore, the null hypothesis of no explosive dynamics could not be rejected for any Magnificent Seven stock in Period 2.

This finding does not mean that the AI-related rally was economically unimportant. Stocks such as Nvidia, Alphabet, and Microsoft experienced strong market attention and substantial appreciation during the post-2022 period. However, the econometric evidence suggests that this appreciation does not satisfy the formal statistical definition of explosive behavior under the GSADF framework used in this thesis. In this extended sample, which includes observations until June 2026, the post-2022 AI rally appears to be strong but non-explosive. One possible explanation is that extending the sample beyond the reference article's January 2025 endpoint captures later market adjustments and valuation normalization, reducing the statistical evidence of exuberance.

The bubble severity analysis further refines these findings. Using the near-unit-root framework of Phillips (P. C. Phillips, 2023), the thesis estimated the autoregressive coefficient, the localizing rate parameter, and the confidence intervals for each stock across the four methodologies. The key conclusion is that the Magnificent Seven stocks are generally characterized by high persistence rather than statistically confirmed Mildly Explosive behavior. Several point estimates are economically interesting. Nvidia appears as a relevant candidate under the Price-to-Index methodology in Period 2, Alphabet shows explosive-side coefficients under some factor specifications, and Tesla records a low localizing rate under the Price-to-Sales specification in Period 1. However, once the confidence intervals are considered, most cases remain statistically compatible with Local-to-Unity behavior.

This distinction is central to the contribution of the thesis. The visual behavior of stock prices, especially during technological revolutions, can easily resemble a speculative bubble. However, the econometric evidence shows that strong growth and high persistence are not the same as statistically confirmed explosive dynam-

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ics. The results indicate that while speculative episodes did occur in the first period, especially for Microsoft and Nvidia, the broader evidence does not support a severe and persistent speculative bubble across all Magnificent Seven stocks. In the second period, the AI-driven recovery appears more consistent with persistent market repricing than with a formally detectable bubble.

Compared with the reference study by Basele, Peter C B Phillips, and Shi (2025), this thesis reaches a more cautious conclusion. The reference article finds stronger evidence of speculative behavior during the recent AI boom, especially in the shorter second sample period. By extending the sample to June 2026, using the S&P 500 as the market benchmark in the Price-to-Index specification, and adding the Fama-French five-factor idiosyncratic component, this thesis applies a broader and stricter empirical design. These methodological choices lead to weaker bubble detection results and more conservative severity classifications.

The main contribution of this thesis is therefore twofold. First, it applies modern real-time bubble detection techniques to the Magnificent Seven using multiple specifications that control for fundamentals, market movements, and systematic risk factors. Second, it goes beyond simple bubble detection by measuring the severity of the estimated dynamics through near-unit-root inference. This allows the analysis to distinguish between Mildly Integrated, Local-to-Unity, and Mildly Explosive regimes, giving a more precise interpretation of speculative risk.

There are also limitations to the analysis. The results are sensitive to sample length, frequency, benchmark selection, and minimum window size. Period 2 remains relatively short for explosive root testing, even though it is longer than the sample used in the reference study. In addition, valuation measures such as the Price-to-Sales ratio cannot capture all aspects of firm fundamentals, especially for companies whose value depends heavily on expected future growth, intangible assets, and technological leadership. The factor models also remove systematic risk, but they cannot perfectly separate rational investor expectations from speculative optimism.

Future research could extend this analysis in several directions. First, the same methodology could be applied again as more post-2026 data become available, allowing a stronger evaluation of whether the AI rally later evolves into an explosive regime. Second, alternative benchmarks such as the Nasdaq-100, S&P 500 Information Technology Index, or equal-weighted market indices could be compared to assess the sensitivity of the Price-to-Index results. Third, future work could include additional firm-level fundamentals such as earnings growth, free cash flow, capital expenditure, and AI-related revenue exposure. Finally, the analysis could be extended to other AI-related companies beyond the Magnificent Seven to test whether speculative dynamics are concentrated only in mega-cap firms or spread more broadly across the technology sector.

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Overall, this thesis finds that the Magnificent Seven experienced episodes of speculative exuberance, particularly during the first period and especially for Microsoft and Nvidia. However, the evidence becomes weaker after applying stricter factor-based controls and after extending the second sample period to June 2026. The post-2022 AI boom shows strong persistence and valuation pressure, but not a statistically confirmed Mildly Explosive bubble under the empirical framework used in this research. The final conclusion is therefore cautious: the Magnificent Seven display signs of speculative pressure, but the evidence does not support the existence of a severe, robust, and persistent asset bubble across the full group.

Appendix A

Data Sources and Variable Construction

This appendix summarizes the main data sources and variable transformations used in the empirical analysis. The construction of the variables follows the objective of isolating speculative price dynamics from fundamental growth, broader market movements, and systematic risk factors.

Table A.1: Data Sources and Variable Construction

Variable	Source	Frequency	Transformation / Use
Adjusted stock prices of the Magnificent Seven	Yahoo Finance	Monthly	End-of-month adjusted closing prices were collected for Alphabet, Amazon, Apple, Meta, Microsoft, Nvidia, and Tesla. Adjusted prices were used in order to account for stock splits and dividend adjustments. These series form the basis for all price-based ratios and return calculations.

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APPENDIX A. DATA SOURCES AND VARIABLE CONSTRUCTION

Table A.1 (continued)

Variable	Source	Frequency	Transformation / Use
Sales / Revenue	Financial statement data provider	Quarterly	Sales data were used to construct the Price-to-Sales ratio. Since sales are reported quarterly while stock prices are observed monthly, the most recent available sales value was carried forward until the next quarterly observation.
Price-to-Sales ratio	Constructed by the author	Monthly	The Price-to-Sales ratio was calculated by dividing each firm's adjusted stock price by its sales measure. This variable was used as a fundamental valuation metric to test whether stock prices increased faster than firm revenues.
S&P 500 Index	Yahoo Finance / market index data	Monthly	The S&P 500 index was used as the benchmark for the Price-to-Index ratio. It was selected because it represents the broader U.S. equity market and is less concentrated in technology stocks than the Nasdaq index.
Price-to-Index ratio	Constructed by the author	Monthly	The Price-to-Index ratio was calculated by dividing each Magnificent Seven stock price by the S&P 500 index. This ratio controls for aggregate market movements and tests whether each stock outperformed the broader market in an explosive way.

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APPENDIX A. DATA SOURCES AND VARIABLE CONSTRUCTION

Table A.1 (continued)

Variable	Source	Frequency	Transformation / Use
Stock returns	Constructed from adjusted stock prices	Monthly	Monthly stock returns were calculated from adjusted prices. These returns were used as the dependent variable in the Fama-French factor regressions.
Risk-free rate	Kenneth French Data Library	Monthly	The risk-free rate was used to compute excess stock returns. It was subtracted from individual stock returns in the Fama-French regressions.
Fama-French three factors	Kenneth French Data Library	Monthly	The three factors include excess market return, SMB, and HML. They were used to estimate the Fama-French three-factor model and extract the idiosyncratic component of each stock.
Fama-French five factors	Kenneth French Data Library	Monthly	The five-factor model extends the three-factor model by adding RMW and CMA. These factors control for profitability and investment effects, which are especially relevant for large technology firms.
FF3 idiosyncratic component	Constructed by the author	Monthly	The residuals from the Fama-French three-factor regression were accumulated over time to construct the FF3 idiosyncratic component. The PSY test was then applied to this component to detect firm-specific speculative behavior after controlling for market, size, and value factors.

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APPENDIX A. DATA SOURCES AND VARIABLE CONSTRUCTION

Table A.1 (continued)

Variable	Source	Frequency	Transformation / Use
FF5 idiosyncratic component	Constructed by the author	Monthly	The residuals from the Fama-French five-factor regression were accumulated over time to construct the FF5 idiosyncratic component. This specification provides the strictest test of idiosyncratic exuberance because it controls for market, size, value, profitability, and investment factors.
Statistical tests	Constructed by the author using R	Monthly	The PSY and GSADF tests were implemented using the <code>exuber</code> package in R. Period 1 was analyzed using the PSY real-time detection procedure, while Period 2 was analyzed using the GSADF statistic with simulated critical values.
Bubble severity measures	Constructed by the author	Monthly	The autoregressive coefficient $\hat{\theta}_n$, the localizing rate parameter $\hat{\gamma}_n$, and the confidence intervals for γ were calculated to classify the dynamics as Mildly Integrated, Local-to-Unity, or Mildly Explosive.

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